

Bernard Birendra Das

Data Engineer

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PROFILE SUMMARY

Data Engineer and PhD Scholar in Data Science with 2+ years of professional software engineering experience designing scalable data architectures, high-throughput streaming pipelines, and microservices. Proficient in Apache Spark, Apache Kafka, PostgreSQL, BigQuery, and containerized cloud environments. Proven success in optimizing query performance, building resilient ETL/ELT workflows, and scaling data platforms for high concurrency.

WORK EXPERIENCE

Simplify3x Software Private Limited • Software Engineer

January 2023 - May 2025 | 2 years 5 months | Bengaluru, Karnataka

- Developed and maintained two impactful modules of SimplifyQA product for the new UI, namely PDF Reports and Pipeline Executions.
- Designed and implemented RESTful APIs and microservices that led to significant improvements in response times and improving overall system scalability.
- Developed dynamic, data-driven dashboards using Angular and D3.js, enabling real-time visualization of key performance metrics for business stakeholders.

PROJECTS

Distributed Real-Time Streaming Pipeline • Apache Kafka, PySpark, PyTorch, Docker, Kubernetes

2026

- Engineered a high-throughput streaming pipeline using Kafka and PySpark to ingest and process user interaction logs.
- Implemented distributed schema transformations and microservice containers managed on Kubernetes for auto-scaling.

Data Pipeline & Execution Engine • PostgreSQL, Node.js, REST APIs, Docker, Java

2024

- Architected resilient execution pipelines and backend microservices handling high-volume parallel test execution records.
- Optimized PostgreSQL queries and database schemas, slashing API response times and boosting system throughput.

EDUCATION

CHRIST (Deemed-to-be-University), Bangalore

PhD in Data Science - 3.6/4.0 GPA

Dec 2024 - Present

MCA with Data Analytics - 8.66 CGPA

June 2021 - June 2023

St. Xavier's College (Autonomous), Kolkata

BSc (Honours) Computer Science - 7.20 CGPA

June 2018 - June 2021

Auxilium Convent School, Barasat

12th Grade, ISC Board - 71.20%

June 2016 - June 2018

10th Grade, ICSE Board - 76.40%

June 2004 - June 2016

SKILLS

PROGRAMMING

Languages:

Python • SQL (PostgreSQL, MySQL) • Java • C++ • Bash

Distributed Processing:

Apache Spark (PySpark) • Apache Kafka • Hadoop • MapReduce

Databases & Warehousing:

PostgreSQL • BigQuery • MongoDB • Redis • SQLite

SYSTEMS & DEVOPS

Containers & Cloud:

Docker • Kubernetes • AWS (S3, EC2) • GCP

Workflow & Tooling:

Airflow • GitHub Actions • Linux/Shell • REST APIs • Git

CORE COMPETENCIES

ETL/ELT Pipeline Design • Stream Processing • Data Modeling • Query Optimization • Distributed Systems

CERTIFICATIONS

Foundational C# with Microsoft

Developer Certification

Sept 2023

AWS Academy Graduate

AWS Academy Cloud Foundations

Nov 2022